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# Name : Richa Patel
# Course Code and Name: 3CS502ME24 OPEN SOURCE TECHNOLOGIES
# Roll no : 24BTM058
# Practical NO : 4
# Aim :Install Jupyter Notebook and Data Visualisation with
Matplotlib: Use Matplotlib to create various data visualisations, such
as bar charts, scatter plots, and histograms.
# Methodology Followed:
```

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("BostonHousing.csv")

print("Basic Information:")
print(df.info())
print(df.describe())
print("\nFirst 5 Rows:")
display(df.head())
```

Basic Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 506 entries, 0 to 505

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	crim	506 non-null	float64
1	zn	506 non-null	float64
2	indus	506 non-null	float64
3	chas	506 non-null	int64
4	nox	506 non-null	float64
5	rm	506 non-null	float64
6	age	506 non-null	float64
7	dis	506 non-null	float64
8	rad	506 non-null	int64
9	tax	506 non-null	int64
10	ptratio	506 non-null	float64
11	b	506 non-null	float64
12	lstat	506 non-null	float64
13	medv	506 non-null	float64

dtypes: float64(11), int64(3)

memory usage: 55.5 KB

None

	crim	zn	indus	chas	nox
rm \					
count	506.000000	506.000000	506.000000	506.000000	506.000000
mean	3.613524	11.363636	11.136779	0.069170	0.554695
std	8.601545	23.322453	6.860353	0.253994	0.115878

```

0.702617
min      0.006320      0.000000      0.460000      0.000000      0.385000
3.561000
25%      0.082045      0.000000      5.190000      0.000000      0.449000
5.885500
50%      0.256510      0.000000      9.690000      0.000000      0.538000
6.208500
75%      3.677083      12.500000      18.100000      0.000000      0.624000
6.623500
max      88.976200     100.000000      27.740000      1.000000      0.871000
8.780000

```

```

          age          dis          rad          tax          ptratio
b \
count  506.000000    506.000000    506.000000    506.000000    506.000000
506.000000
mean    68.574901      3.795043      9.549407    408.237154    18.455534
356.674032
std     28.148861      2.105710      8.707259    168.537116     2.164946
91.294864
min      2.900000      1.129600      1.000000    187.000000    12.600000
0.320000
25%     45.025000      2.100175      4.000000    279.000000    17.400000
375.377500
50%     77.500000      3.207450      5.000000    330.000000    19.050000
391.440000
75%     94.075000      5.188425      24.000000    666.000000    20.200000
396.225000
max     100.000000     12.126500     24.000000    711.000000    22.000000
396.900000

```

```

          lstat          medv
count  506.000000    506.000000
mean    12.653063     22.532806
std      7.141062      9.197104
min      1.730000      5.000000
25%      6.950000     17.025000
50%     11.360000     21.200000
75%     16.955000     25.000000
max     37.970000     50.000000

```

First 5 Rows:

```

          crim      zn  indus  chas      nox      rm      age      dis      rad      tax
ptratio \
0  0.00632    18.0    2.31      0    0.538    6.575    65.2    4.0900      1    296
15.3
1  0.02731      0.0    7.07      0    0.469    6.421    78.9    4.9671      2    242
17.8
2  0.02729      0.0    7.07      0    0.469    7.185    61.1    4.9671      2    242

```

```

17.8
3 0.03237 0.0 2.18 0 0.458 6.998 45.8 6.0622 3 222
18.7
4 0.06905 0.0 2.18 0 0.458 7.147 54.2 6.0622 3 222
18.7

```

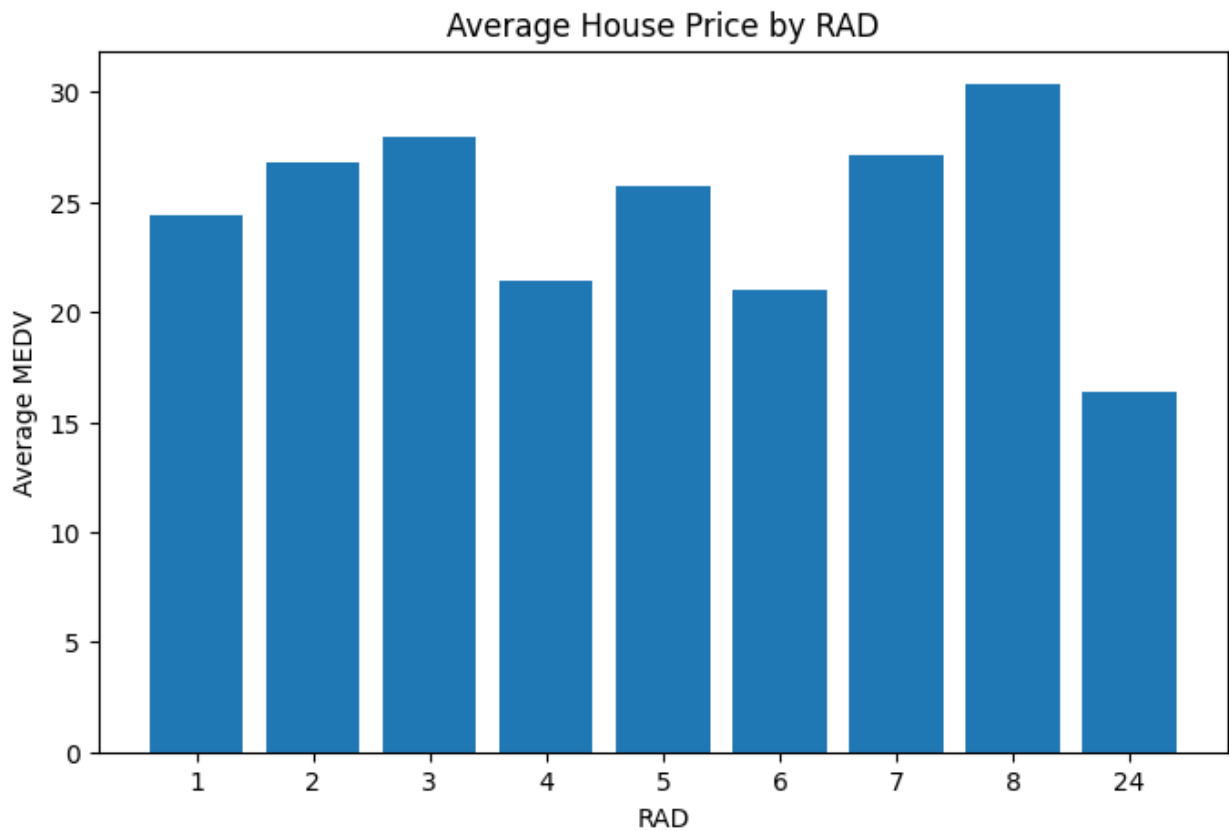
	b	lstat	medv
0	396.90	4.98	24.0
1	396.90	9.14	21.6
2	392.83	4.03	34.7
3	394.63	2.94	33.4
4	396.90	5.33	36.2

```

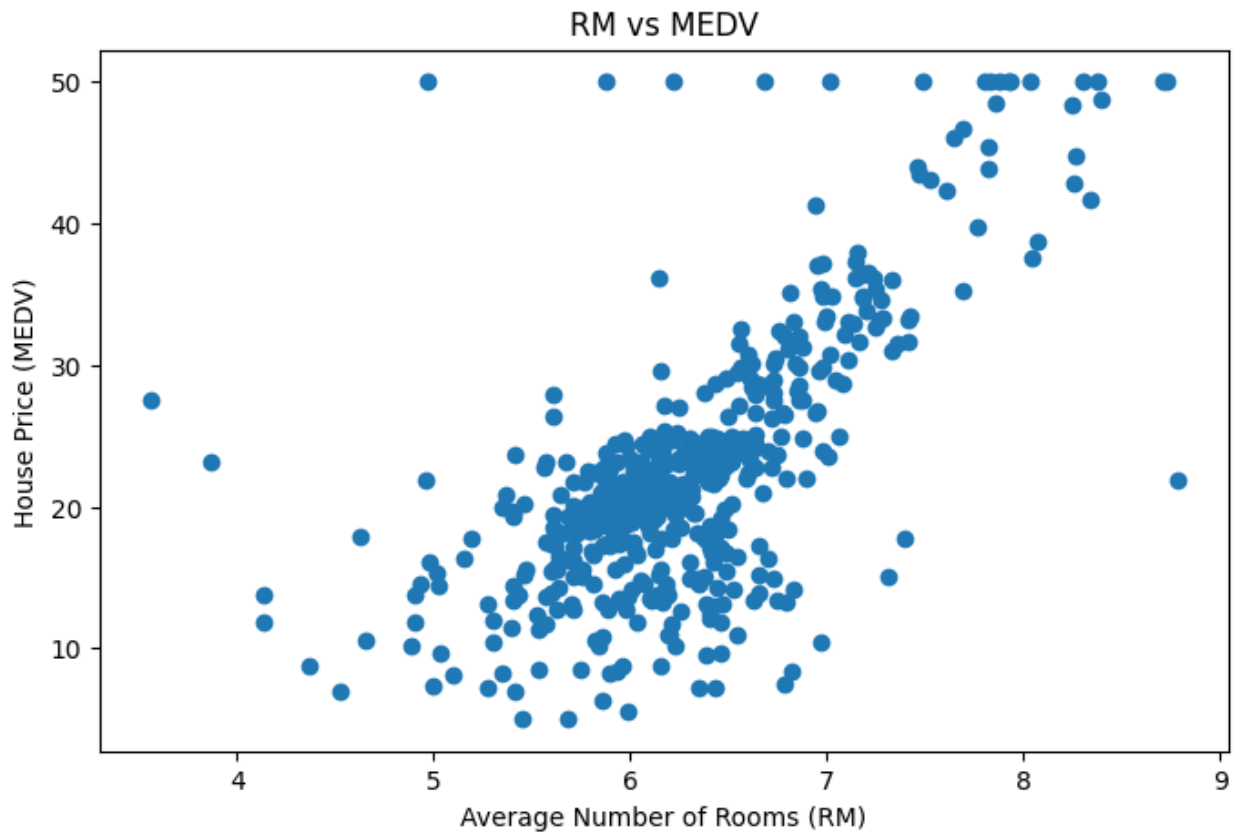
avg_medv = df.groupby("rad")["medv"].mean()

plt.figure(figsize=(8, 5))
plt.bar(avg_medv.index.astype(str), avg_medv.values)
plt.xlabel("RAD")
plt.ylabel("Average MEDV")
plt.title("Average House Price by RAD")
plt.show()

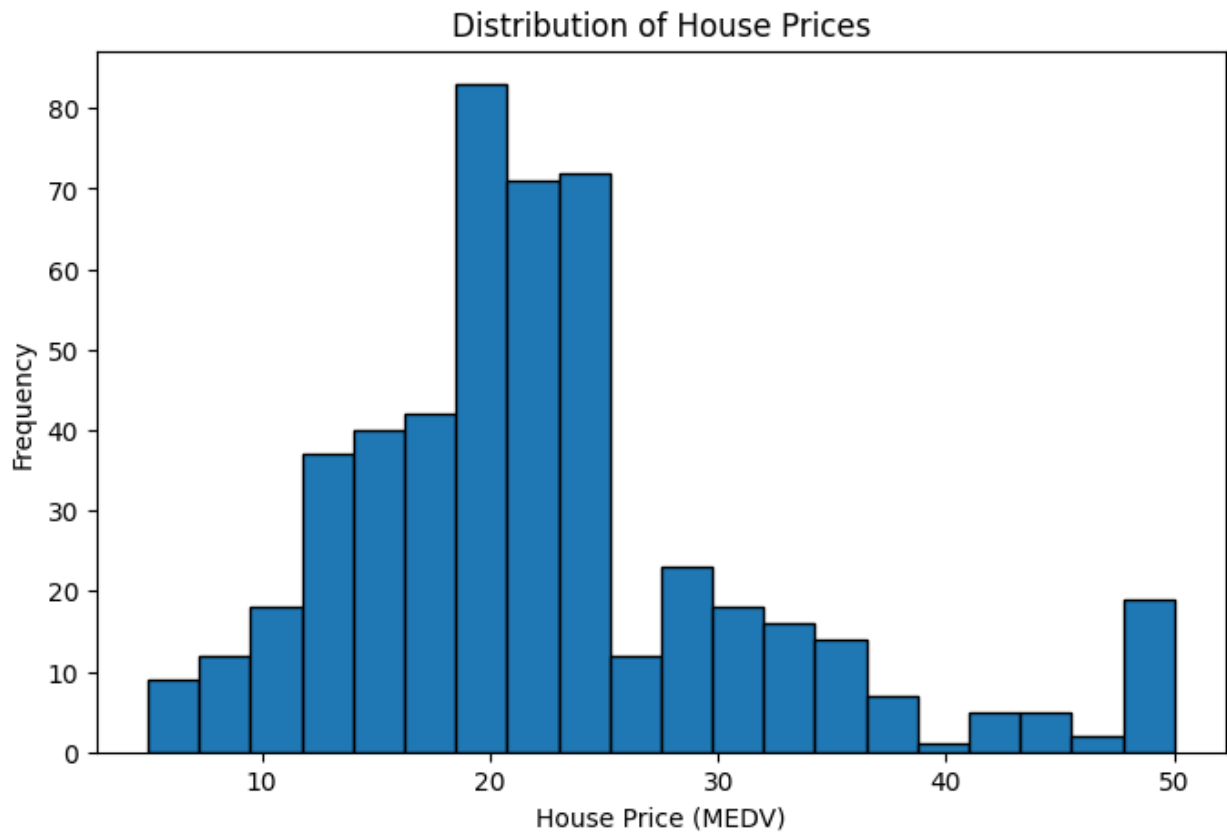
```



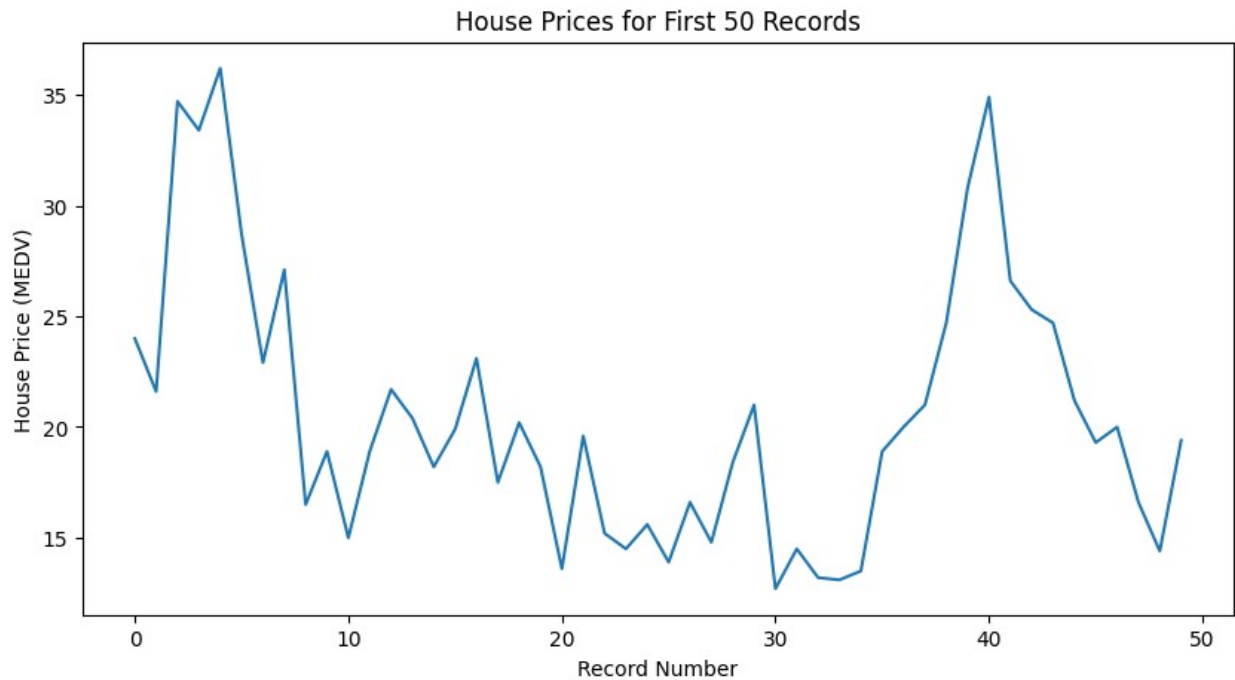
```
plt.figure(figsize=(8, 5))
plt.scatter(df["rm"], df["medv"])
plt.xlabel("Average Number of Rooms (RM)")
plt.ylabel("House Price (MEDV)")
plt.title("RM vs MEDV")
plt.show()
```



```
plt.figure(figsize=(8, 5))
plt.hist(df["medv"], bins=20, edgecolor="black")
plt.xlabel("House Price (MEDV)")
plt.ylabel("Frequency")
plt.title("Distribution of House Prices")
plt.show()
```



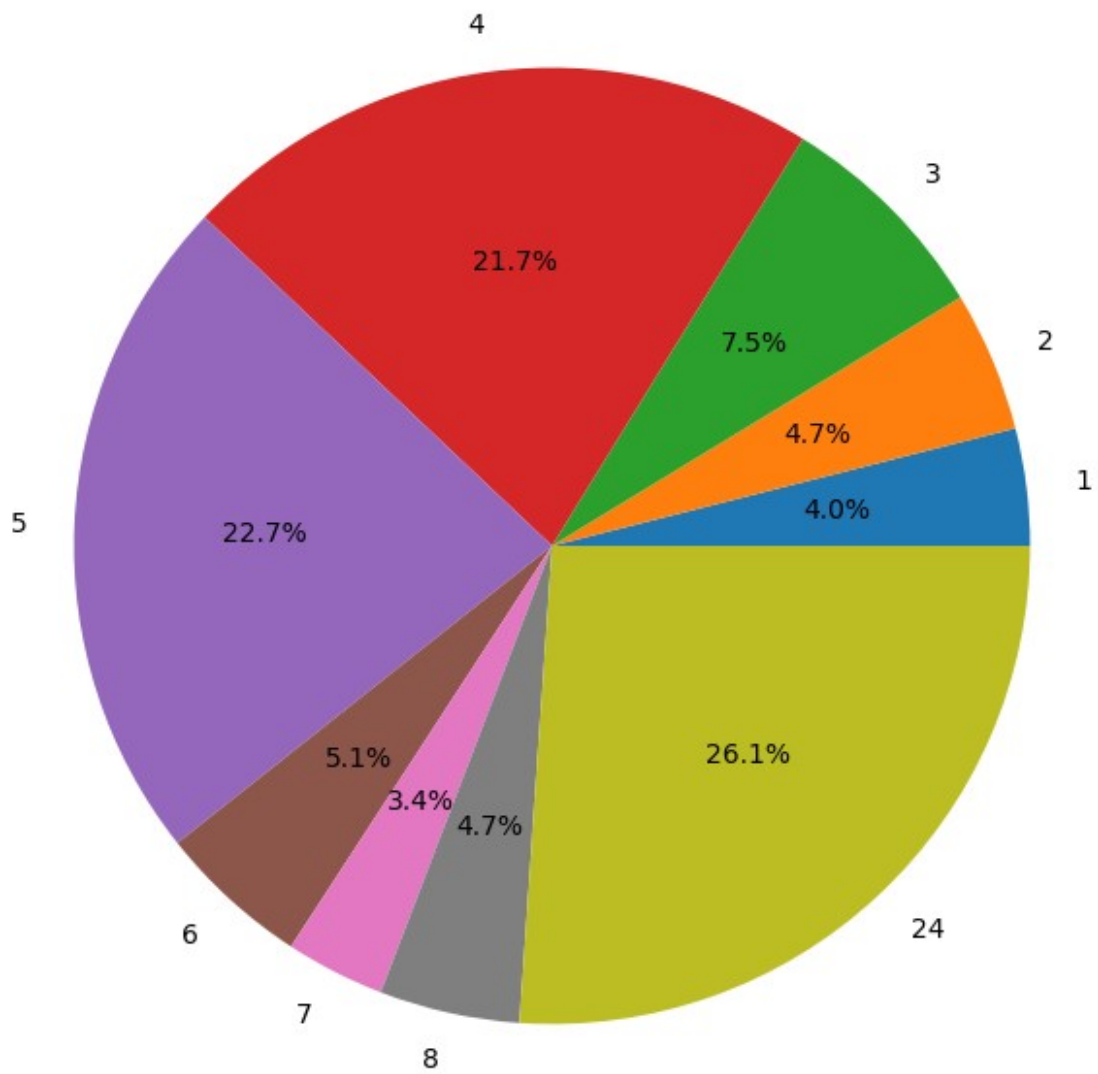
```
plt.figure(figsize=(10, 5))
plt.plot(df.index[:50], df["medv"][:50])
plt.xlabel("Record Number")
plt.ylabel("House Price (MEDV)")
plt.title("House Prices for First 50 Records")
plt.show()
```



```
rad_counts = df["rad"].value_counts().sort_index()

plt.figure(figsize=(8, 8))
plt.pie(
    rad_counts.values,
    labels=rad_counts.index,
    autopct="%1.1f%%"
)
plt.title("Distribution of Houses by RAD")
plt.show()
```

Distribution of Houses by RAD



```
plt.figure(figsize=(8, 5))
plt.boxplot(df["medv"])
plt.ylabel("House Price (MEDV)")
plt.title("Box Plot of House Prices")
plt.show()
```

Box Plot of House Prices

